

Embedded Memory Choices For 65nm SoCs

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Agenda

- Application segments and SoC architectures
- Embedded memory portfolio
- Appropriate memory for your architecture
- Summary

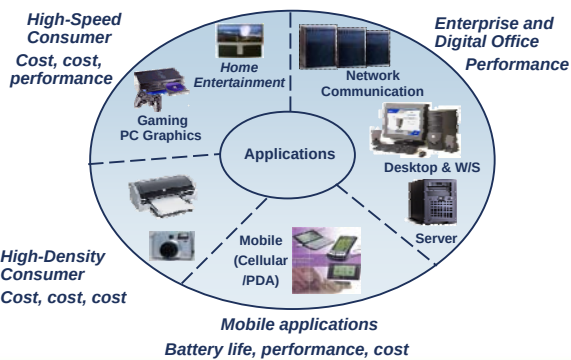


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SoC Application Segments



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Four Categories of SoC Design

- Enterprise and digital office
 - High speed required over the entire chip (>750MHz)
 - Typically Use G or high-speed processes
- Mobile applications
 - High speed required for embedded processor (~650MHz)
 - High density for rest of the SoC (~300MHz)
 - Typically use low leakage processes
- High-speed consumer
 - High speed required for embedded processor (~650MHz)
 - High density for rest of the SoC (~300MHz)
 - Could use either low leakage or generic processes
- High-density consumer
 - Moderate to low speed required over the entire chip (<~300MHz)
 - High density required over the entire chip
 - Typically use low leakage processes



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Enterprise and Digital Office Segment

- High speed required over the entire chip (>750MHz)
- Typically use G or high-speed processes
- Speed is the key criterion
 - Setup time + access time
 - Memories need to support pipelined outputs for better timing
- High-capacity memories are also required
 - Up to 2Mbits of single- and dual-port SRAM



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Mobile Applications Segment

- High speed required for embedded processor (~650MHz)
- High density for rest of the SoC (~300MHz)
- Aggressive power management
 - Low leakage "LP/LL" processes
 - Multi-VT designs
 - Adaptive voltage scaling
 - Low voltage operation
 - Retention and shutdown modes



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ARM

High-Speed Consumer Segment

- High speed required for embedded processor (~650MHz)
- High density for rest of the SoC (~300MHz)
- Moderate power management
 - G or Low leakage "LP/LL" processes
 - Multi-VT designs
 - Voltage islands
- High-capacity memories may be required
 - Up to 4Mbits of single-port SRAM



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ARM

High-Density Consumer Segment

- Moderate speed required over entire SoC (<300 MHz)
- High density required for entire SoC
- Moderate power management
 - Low leakage "LP/LL" processes
 - Multi-VT designs
 - Voltage islands
- Sub-segment of low speed SoCs (< 100MHz)
 - Very low leakage requirements
 - Low voltage operation requirements



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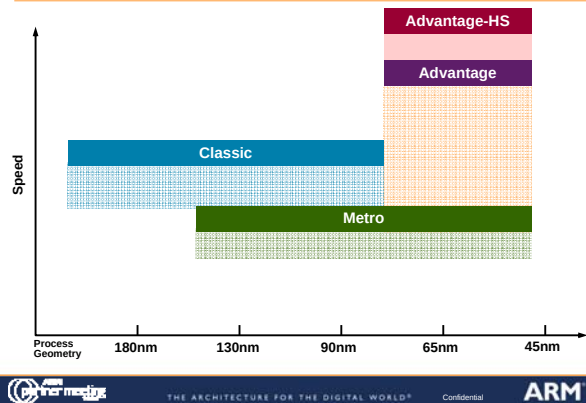


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Memory Platform Solutions



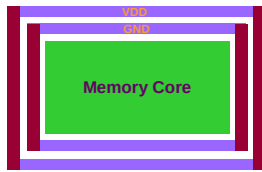
Memory Generator Portfolio

Generator	Range	Ports	Note
High Speed			
Advantage-HS Single Port SRAM	256 – 1+Mbits	1 R/W Port	High Speed
Advantage Single Port SRAM	256 – 1+Mbits	1 R/W Port	High Performance
Advantage Dual Port SRAM	128 - 256Kbits	2 R/W Ports	High Performance
Advantage Single Port Register File	64 - 32Kbits	1 R/W Port	High Performance
Advantage Two Port Register File	16 - 64Kbits	1 R Port, 1 W Port	High Performance
Advantage Via Programmable ROM	128 - 1Mbits	1 R Port	High Performance
High Density			
Metro Ultra High Density Two Port Register File	16 - 16Kbits	1 R Port, 1 W Port	High Density
Metro Single Port SRAM	256 – 1+Mbits	1 R/W Port	High Density
Metro Dual Port SRAM	128 - 256Kbits	2 R/W Ports	High Density
Metro Via Programmable ROM	128 - 1Mbits	1 R Port	High Density

Advantage90 Memory Features

- All Classic features
- High-performance architecture
- Extra margin adjustment
 - Multiple choices of internal timing pulse width
 - Allows post silicon timing adjustment to improve yield
- Flex-Repair™ redundancy
 - One redundant I/O for register files
 - Up to two redundant columns and two redundant rows for SRAMs
- ECC / soft error repair
 - RTL for modified Hamming code
 - Extra bits for storing code
- Integrated BIST test muxes
- Choice of power rings or ring-less power routing

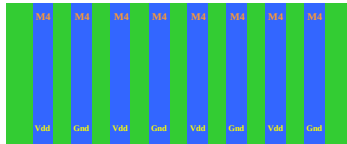
Power Rings vs ArtiGrid™ Ringless



Power rings

- Wider usage due to traditional style
- Sizable overhead in case of large number of small memories

ArtiGrid For SRAM



ArtiGrid ringless routing

- Significant area savings in case of a large number of small memories
- Easily routable with industry standard tools



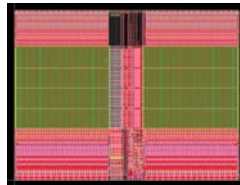
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Advantage65 Memory: Power

- Advantage90 features
- High-performance architecture
 - Optional pipelined outputs
- Optional power down modes
 - Standby
 - Retention
 - Shutdown
- Multi-V_T design for leakage reduction
 - No reduction in performance to achieve low leakage
- Low voltage operation
 - Integrated low-footprint level shifters between core and periphery
 - Enables periphery to run at lower voltage than core array
 - Lowers minimum leakage of the memory instance
 - Innovative write-assist technology to allow stable low voltage operation



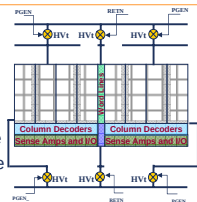
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Memory Power Management Modes

- Standby mode
 - CEN disables the memory
 - Leakage only standby current
- Retention mode
 - Power is supplied to core array to retain state
 - Power is off for periphery for reduced leakage
 - Outputs are clamped to zero
- Shutdown mode
 - Power is off for core and periphery for reduced leakage
 - Outputs are clamped to zero
- Possible through both integrated MT-CMOS as well as separated power sources for core and periphery



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Advantage65 Memory: DFT Features

- Flex-Repair™ redundancy
 - One redundant column for register files
 - Up to two redundant columns and four redundant rows for SRAMs
- Weak bit test
 - Modifies width of internal self timing pulse
 - Stresses read and write operations
- Read disturb test
 - Stresses the read operation to check for bitcell read stability
- Optional BIST and BISR wrappers
 - Works seamlessly with the ARM memory test and repair system
 - Interfaces well with industry standard test tools



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Metro65 Memory Features

- Advantage65 features
- High-density and lower leakage architecture
- Optional power down modes
 - Standby, retention and shutdown
- Dual VT design for leakage reduction
 - Much higher use of high VT devices in the periphery
- Low voltage operation
 - Integrated low-footprint level shifters between core and periphery
- Flex-Repair™ redundancy
 - One redundant I/O for register files
 - Up to two redundant I/O (logical column) and four redundant rows for SRAMs
- Weak bit test
 - Modifies width of internal self timing pulse to stress read and write operations
- Read disturb test
- Optional BIST and BISR wrappers



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Enterprise and Digital Office Segment

- High speed required over the entire chip ($>750\text{MHz}$)
- Typically use G or high speed processes
- Speed is the key criterion
 - Setup time + access time
 - Memories need to support pipelined outputs for better timing
- High-capacity memories are also required
 - Up to 2Mbits of single- and dual-port SRAM
- *Advantage-HS™ memory with pipelined outputs and regular V_T periphery is the most appropriate choice*
- *In some cases, low V_T devices may be used in the periphery to further improve access time*



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Mobile Applications Segment

- High speed required for embedded processor ($\sim 650\text{MHz}$)
- High density for rest of the SoC ($\sim 300\text{MHz}$)
- Aggressive power management
 - Low leakage "LP/LL" processes
 - Multi- V_T designs
 - Adaptive voltage scaling
 - Low voltage operation
 - Retention and shutdown modes
- *Advantage memory with mixed V_T periphery is the most appropriate choice for the high-speed section*
- *Metro™ memory with mixed V_T periphery is the most appropriate choice for the high density section*
- *Advantage and Metro memories support power management modes*



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ARM

High-Speed Consumer Segment

- High speed required for embedded processor ($\sim 650\text{MHz}$)
- High density for rest of the SoC ($\sim 300\text{MHz}$)
- Moderate power management
 - G or Low leakage "LP/LL" processes
 - Multi- V_T designs
 - Voltage islands
- High-capacity memories may be required
 - Up to 4Mbits of single-port SRAM
- *Advantage memory with mixed V_T periphery is the most appropriate choice for the high-speed section*
- *Metro memory with mixed V_T periphery is the most appropriate choice for the high-density section*
- *Memory power management can be enabled where needed*



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ARM

High-Density Consumer Segment

- Moderate speed required over entire SoC (<300 MHz)
- High density required for entire SoC
- Moderate power management
 - Low leakage "LP/LL" processes
 - Multi- V_T designs
 - Voltage islands
- Low-speed sub-segment (< 100MHz)
 - Very low leakage requirements
 - Low voltage operation
- *Metro memory with mixed V_T periphery is the most appropriate choice for the moderate speed segment*
- *Metro memory with all high V_T periphery is the most appropriate choice for the low speed segment*
- *Memory power management should be used across the chip*



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Contents

- Application Segments and SoC Architectures
- ARM Embedded Memory Portfolio
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Summary

- The end application drives the SoC architecture
- Each SoC architecture has specific embedded memory requirements
- Different combinations of performance, power and area become decision criteria
- ARM has a wide portfolio of embedded memory solutions across multiple foundries and nodes
- ARM has the memory solution you need



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